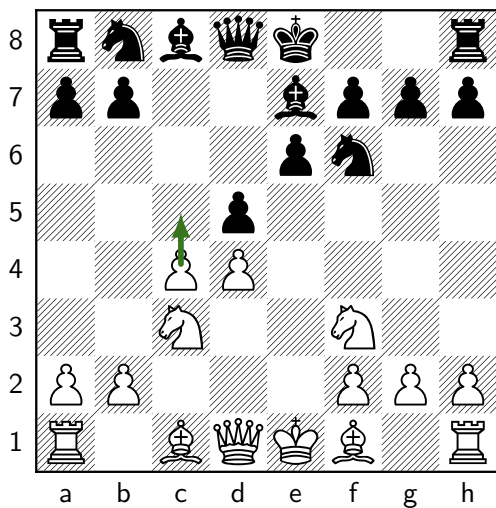


Melanie-XRAY — Caro-Kann, Panov

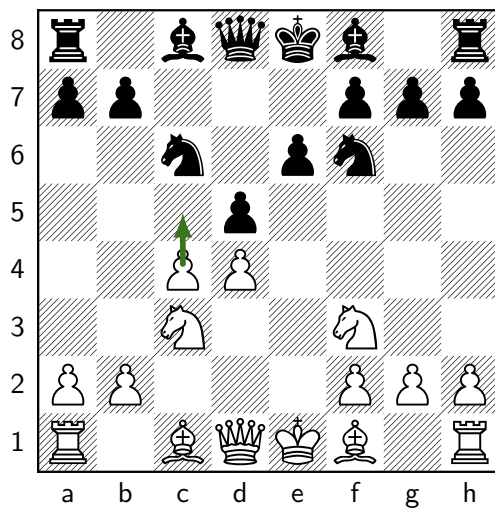
Melanie (Deep_Vision)

Abstract

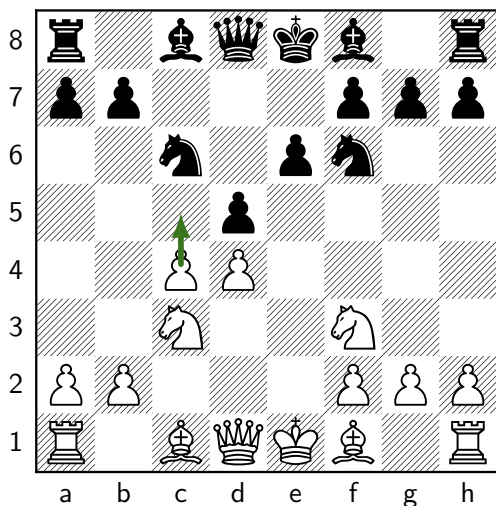
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



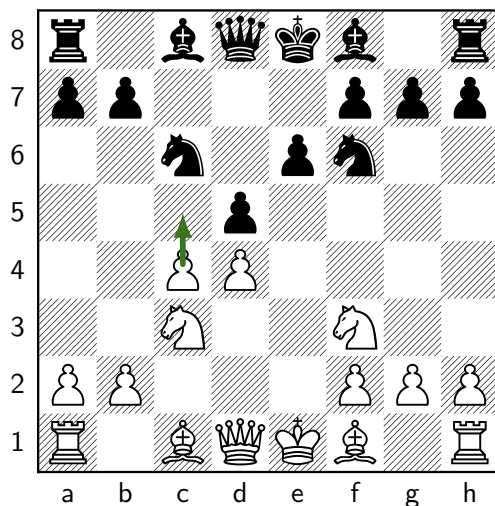
(a)



(b)

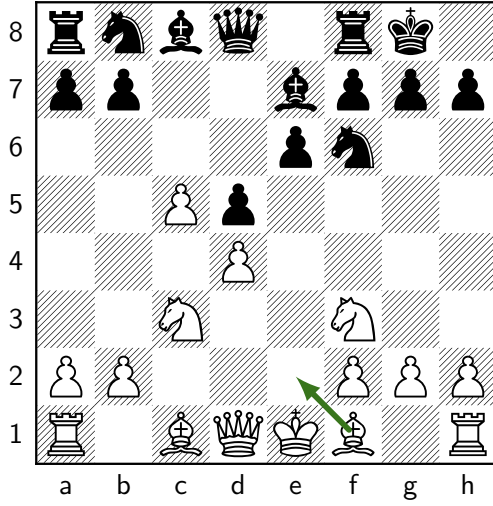


(c)

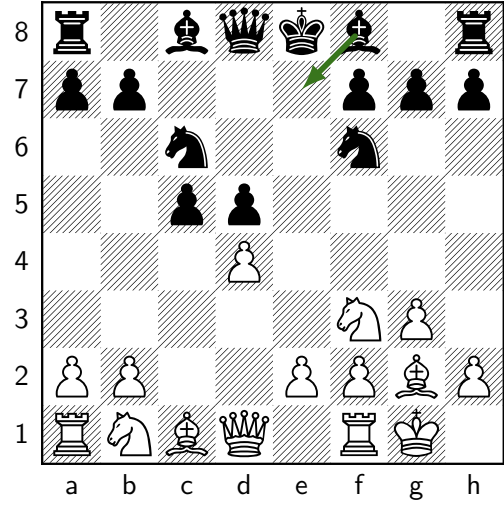


(d)

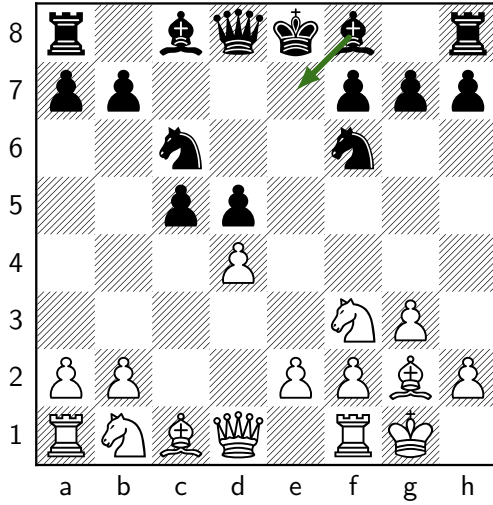
Figure 1: **Move 7, White: c5.** (a) Our position; Melanie plays c5 (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3405` (ply 12) [structural distance $d=29$, QAP obj=115]: the master played c5; under μ our Pc_4 plays the role of that game’s Pc_4 ; cross-type roles: our be_7 as their Pb_2 , our Pd_4 as their Nf_3 ; transferred move c4c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3406` (ply 12) [structural distance $d=29$, QAP obj=115]: the master played c5; under μ our Pc_4 plays the role of that game’s Pc_4 ; cross-type roles: our be_7 as their Pb_2 , our Pd_4 as their Nf_3 ; transferred move c4c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#180` (ply 12) [structural distance $d=29$, QAP obj=115]: the master played c5; under μ our Pc_4 plays the role of that game’s Pc_4 ; cross-type roles: our be_7 as their Pb_2 , our Pd_4 as their Nf_3 ; transferred move c4c5 (**endorses this move**). Synthesis: 21 retrieved analogs transferred to c5 (summed QAP objective 1973, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



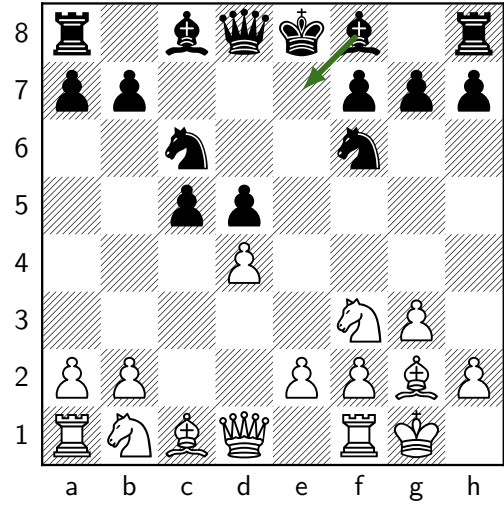
(a)



(b)

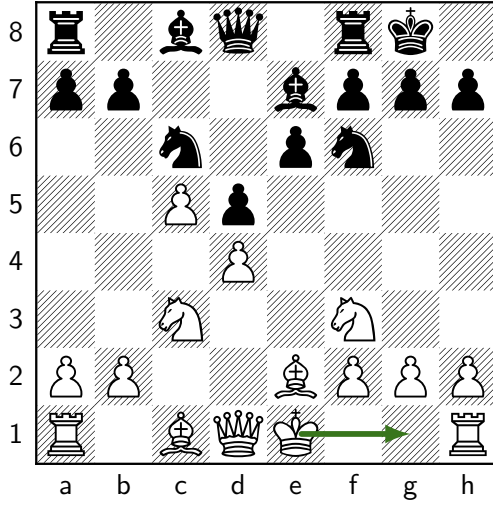


(c)

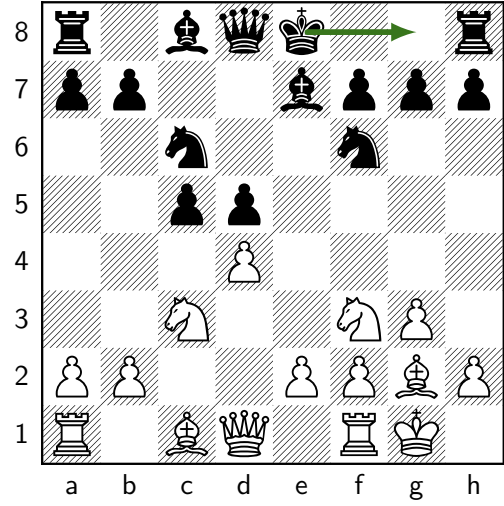


(d)

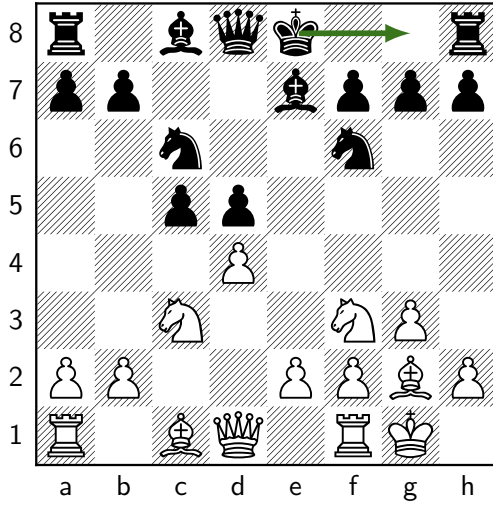
Figure 2: **Move 8, White: Be2**. (a) Our position; Melanie plays **Be2** (**f1e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1124` (ply 13) [structural distance $d=30$, QAP obj=43]: the master played **Be7**; under μ our *Bf1* plays the role of that game’s *Bf1*; cross-type roles: our *pa7* as their *Bc8*, our *Nc3* as their *Pc5*; transferred move **f1e2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1125` (ply 13) [structural distance $d=30$, QAP obj=43]: the master played **Be7**; under μ our *Bf1* plays the role of that game’s *Bf1*; cross-type roles: our *pa7* as their *Bc8*, our *Nc3* as their *Pc5*; transferred move **f1e2** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#959` (ply 13) [structural distance $d=30$, QAP obj=43]: the master played **Be7**; under μ our *Bf1* plays the role of that game’s *Bf1*; cross-type roles: our *pa7* as their *Bc8*, our *Nc3* as their *Pc5*; transferred move **f1e2** (**endorses this move**). Synthesis: 27 retrieved analogs transferred to **Be2** (summed QAP objective 1019, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



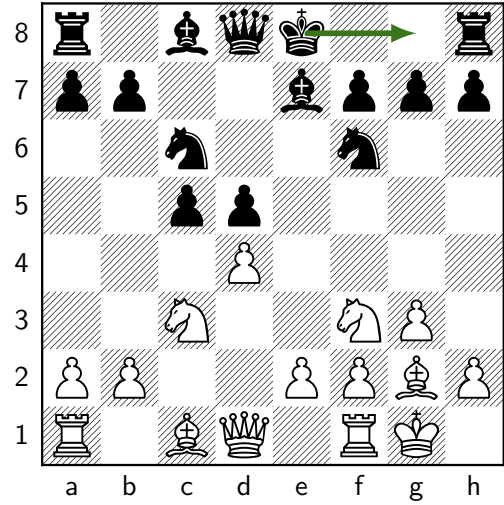
(a)



(b)

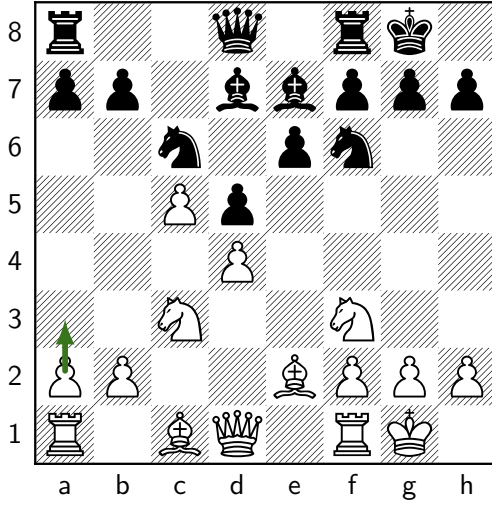


(c)

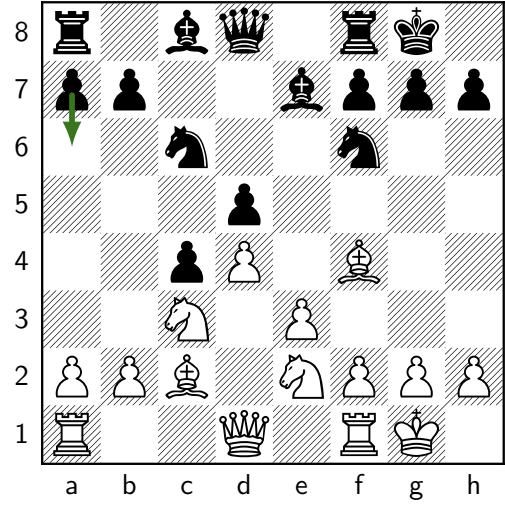


(d)

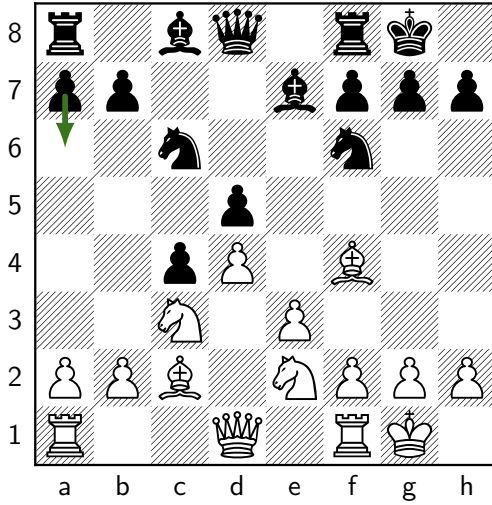
Figure 3: **Move 9, White: 0-0.** (a) Our position; Melanie plays 0-0 (e1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#101` (ply 15) [structural distance $d=29$, QAP obj=61]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ph7*, our *Nc3* as their *Pc5*; transferred move e1g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#102` (ply 15) [structural distance $d=29$, QAP obj=61]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ph7*, our *Nc3* as their *Pc5*; transferred move e1g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#174` (ply 15) [structural distance $d=29$, QAP obj=61]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ph7*, our *Nc3* as their *Pc5*; transferred move e1g1 (**endorses this move**). Synthesis: 39 retrieved analogs transferred to 0-0 (summed QAP objective 2283, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



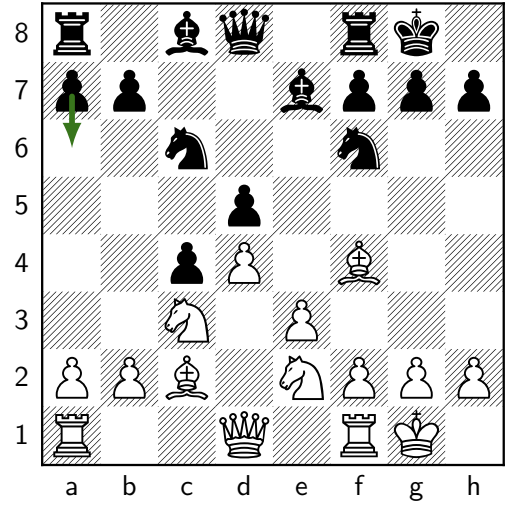
(a)



(b)

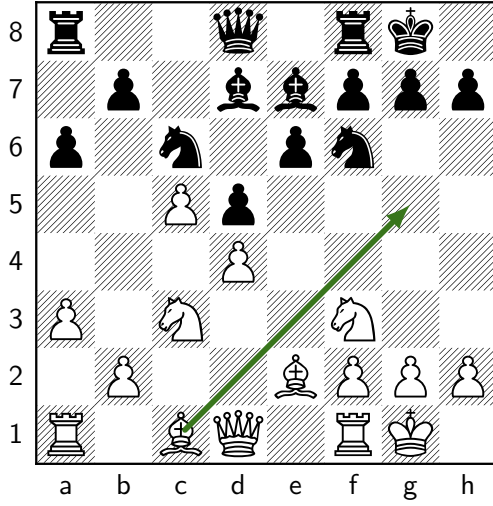


(c)

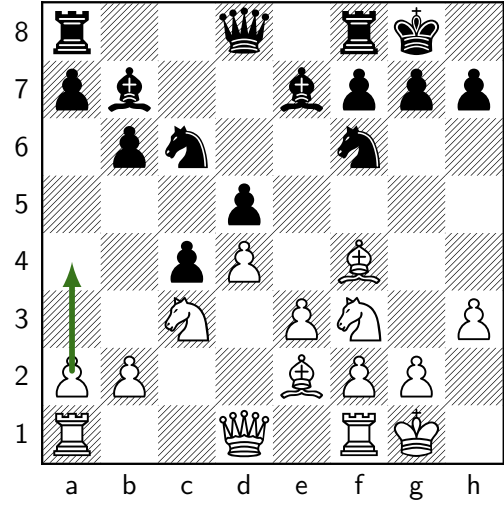


(d)

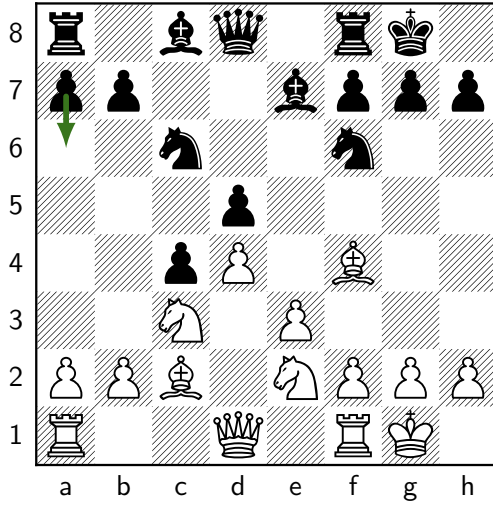
Figure 4: **Move 10, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1586` (ply 19) [structural distance $d=40$, QAP obj=68]: the master played a6; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *nf6* as their *bf4*, our *be7* as their *pa2*; transferred move a2a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2787` (ply 19) [structural distance $d=40$, QAP obj=68]: the master played a6; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *nf6* as their *bf4*, our *be7* as their *pa2*; transferred move a2a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2874` (ply 19) [structural distance $d=40$, QAP obj=68]: the master played a6; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *nf6* as their *bf4*, our *be7* as their *pa2*; transferred move a2a3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to a3 (summed QAP objective 370, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



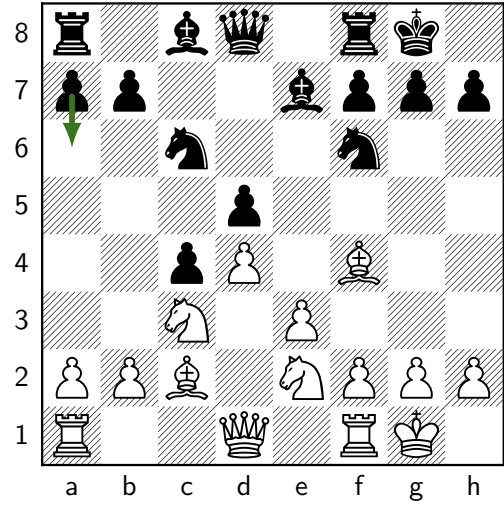
(a)



(b)

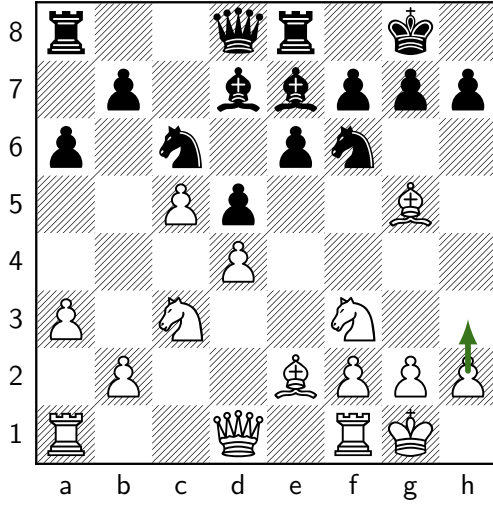


(c)

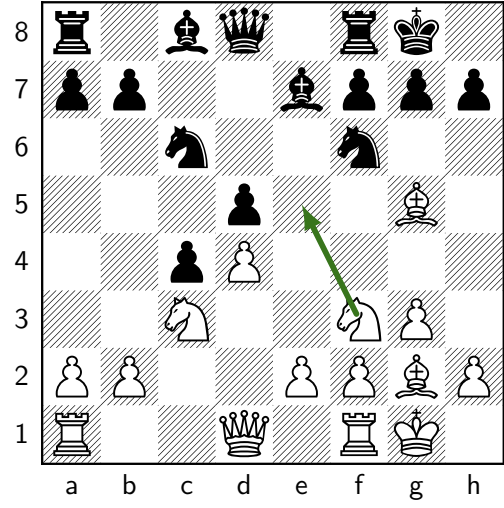


(d)

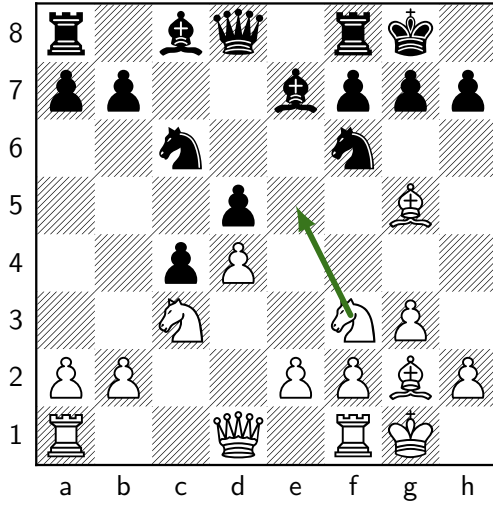
Figure 5: **Move 11, White: Bg5.** (a) Our position; Melanie plays Bg5 (c1g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2772` (ply 24) [structural distance $d=47$, QAP obj=79]: the master played a4; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Pb2* as their *Ra1*, our *Be2* as their *Pb2*; transferred move c1g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1586` (ply 19) [structural distance $d=51$, QAP obj=66]: the master played a6; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Pc5* as their *Bc8*, our *qd8* as their *Pb7*; transferred move c1g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2787` (ply 19) [structural distance $d=51$, QAP obj=66]: the master played a6; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Pc5* as their *Bc8*, our *qd8* as their *Pb7*; transferred move c1g5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bg5 (summed QAP objective 438, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



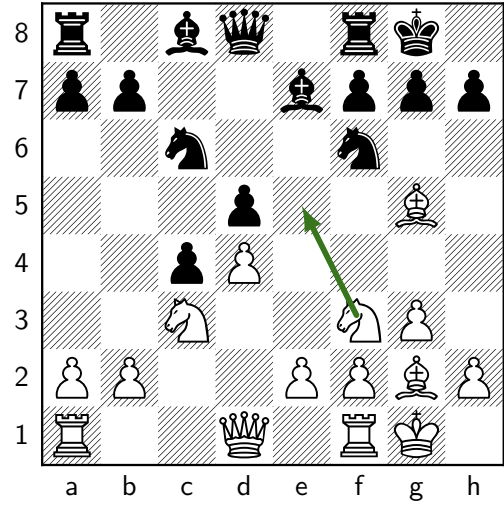
(a)



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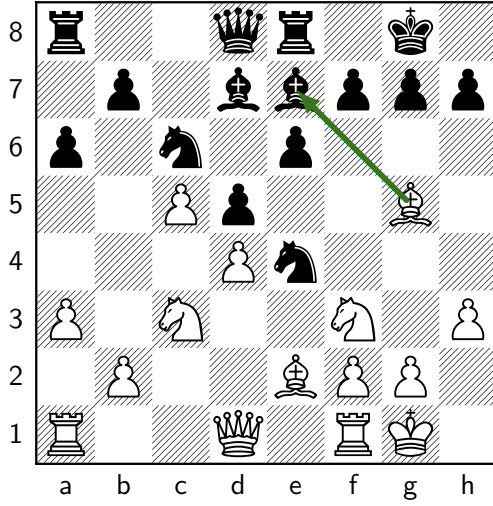


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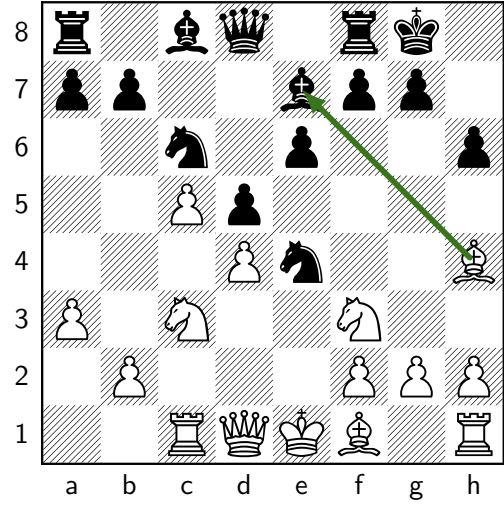


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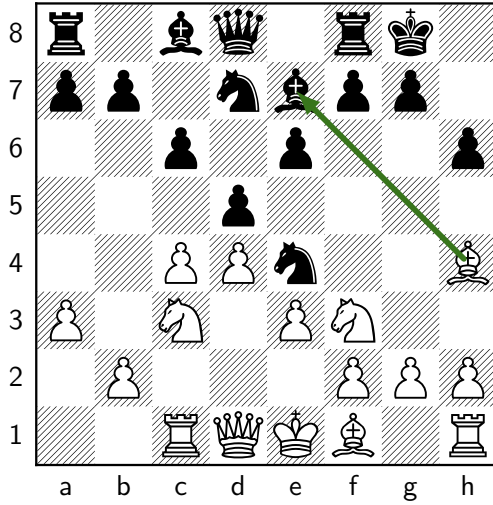
Figure 6: **Move 12, White: h3**. (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#329` (ply 18) [structural distance $d=63$, QAP obj=47]: the master played Ne5; under μ our Ph2 plays the role of that game's Nf3 (a **cross-type** correspondence); cross-type roles: our nc6 as their Pe2, our Pd4 as their Bg2; transferred move h2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2183` (ply 18) [structural distance $d=63$, QAP obj=47]: the master played Ne5; under μ our Ph2 plays the role of that game's Nf3 (a **cross-type** correspondence); cross-type roles: our nc6 as their Pe2, our Pd4 as their Bg2; transferred move h2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2663` (ply 18) [structural distance $d=63$, QAP obj=47]: the master played Ne5; under μ our Ph2 plays the role of that game's Nf3 (a **cross-type** correspondence); cross-type roles: our nc6 as their Pe2, our Pd4 as their Bg2; transferred move h2h3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to h3 (summed QAP objective 268, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



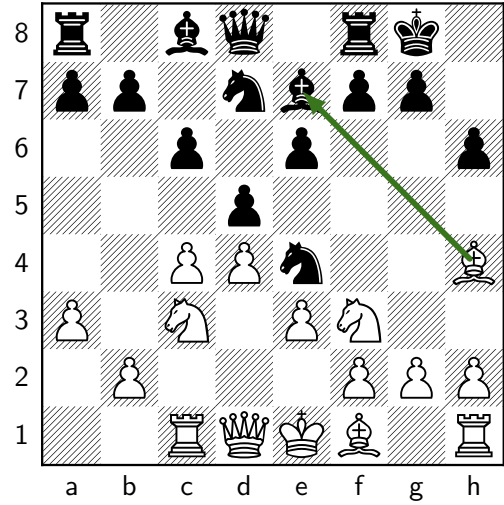
(a)



(b)

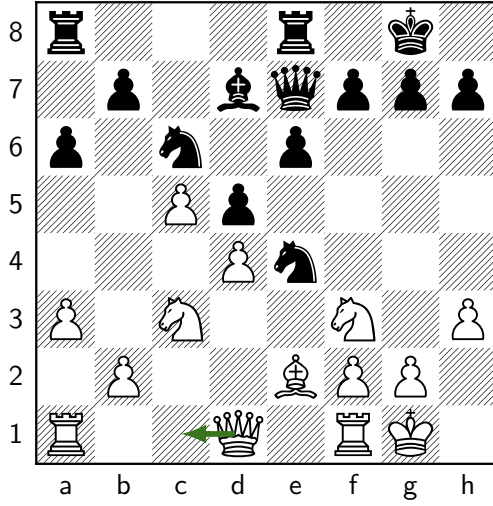


(c)

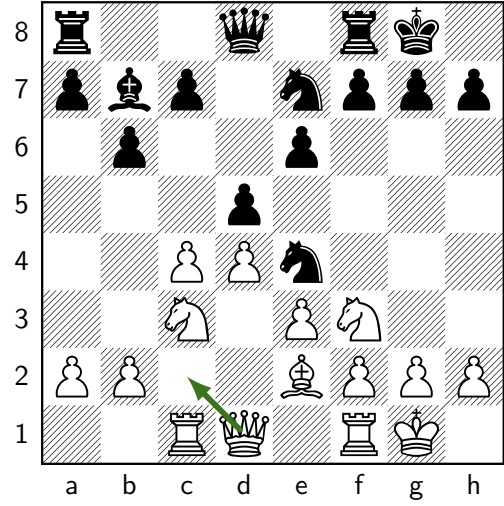


(d)

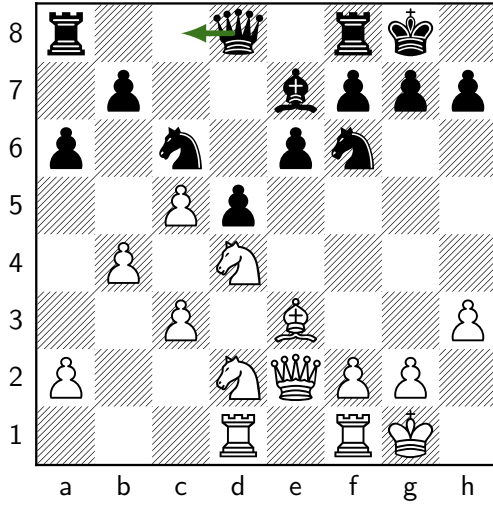
Figure 7: **Move 13, White: Bxe7.** (a) Our position; Melanie plays Bxe7 (g5e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#63` (ply 22) [structural distance $d=67$, QAP obj=191]: the master played Bxe7; under μ our Bg5 plays the role of that game’s Bh4; cross-type roles: our Ph3 as their Bf1, our bd7 as their pb7; transferred move g5e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3316` (ply 18) [structural distance $d=100$, QAP obj=160]: the master played Bxe7; under μ our Bg5 plays the role of that game’s Bh4; cross-type roles: our bd7 as their Rc1, our Rf1 as their Qd1; transferred move g5e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3317` (ply 18) [structural distance $d=100$, QAP obj=160]: the master played Bxe7; under μ our Bg5 plays the role of that game’s Bh4; cross-type roles: our bd7 as their Rc1, our Rf1 as their Qd1; transferred move g5e7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bxe7 (summed QAP objective 1025, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



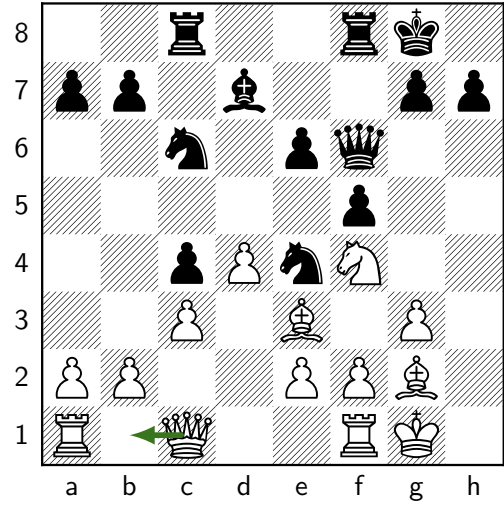
(a)



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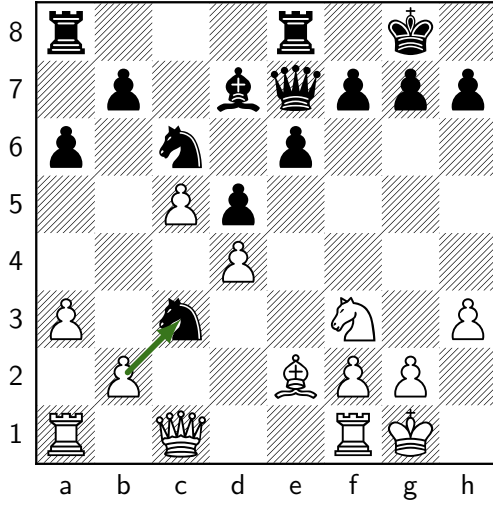


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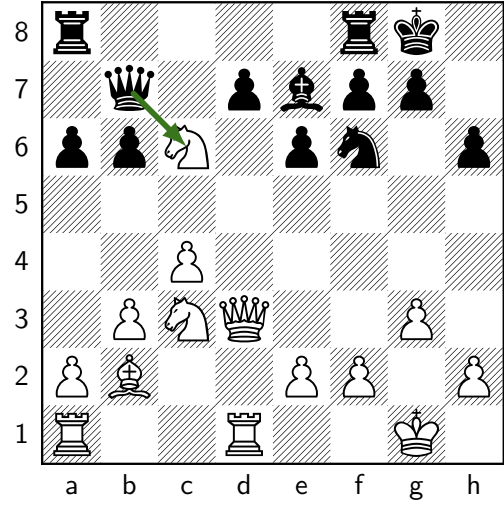


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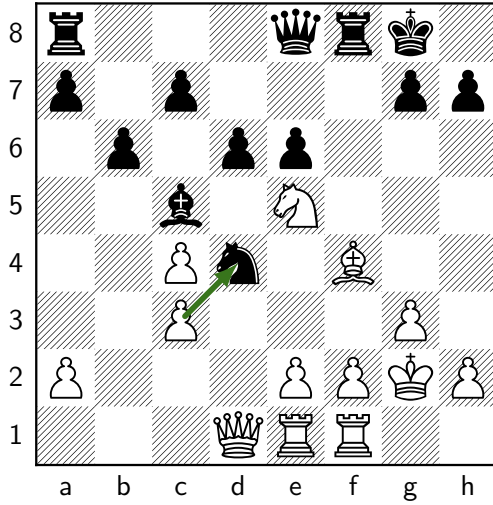
Figure 8: **Move 14, White: Qc1.** (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#295` (ply 20) [structural distance $d=84$, QAP obj=104]: the master played Qc2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Be2 as their Rc1, our Ra1 as their Pa2; transferred move d1c1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#788` (ply 29) [structural distance $d=74$, QAP obj=81]: the master played Qc8; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pe6 as their Ra8, our nc6 as their Be7; transferred move d1c1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#573` (ply 32) [structural distance $d=72$, QAP obj=71]: the master played Qb1; under μ our Qd1 plays the role of that game’s Qc1; cross-type roles: our ne4 as their Bg2, our ne4 as their Pc3; transferred move d1c1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qc1 (summed QAP objective 359, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



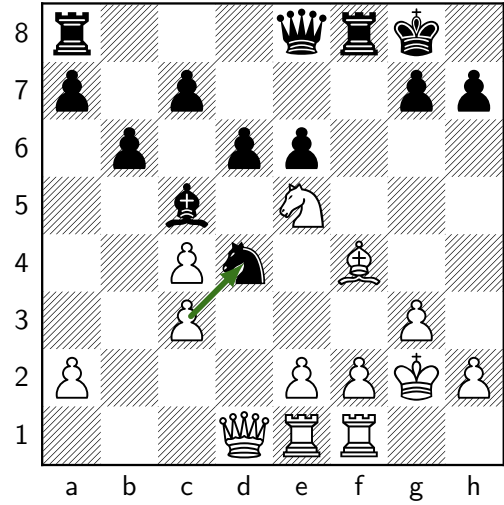
(a)



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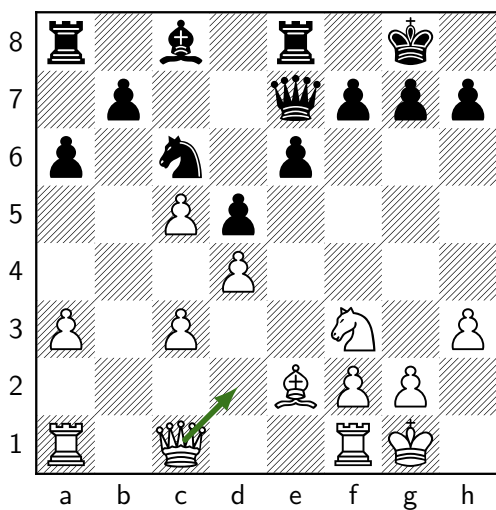


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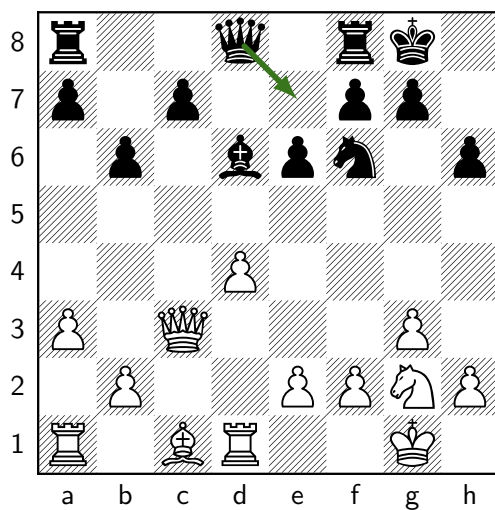


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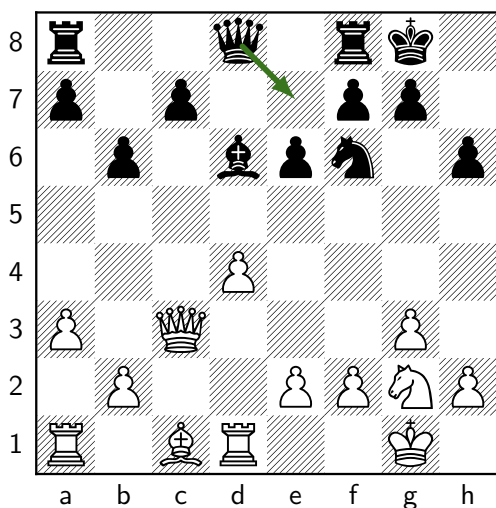
Figure 9: **Move 15, White: bxc3.** (a) Our position; Melanie plays **bxc3** (**b2c3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1160` (ply 29) [structural distance $d=49$, QAP obj=71]: the master played **Qxc6**; under μ our *Pb2* plays the role of that game’s *Qb2* (a **cross-type** correspondence); cross-type roles: our *Qc1* as their *Ra8*, our *Ra1* as their *Pa6*; transferred move **b2c3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1303` (ply 34) [structural distance $d=66$, QAP obj=62]: the master played **cxd4**; under μ our *Pb2* plays the role of that game’s *Pc3*; cross-type roles: our *Pa3* as their *Rf1*, our *Be2* as their *Pe2*; transferred move **b2c3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3252` (ply 34) [structural distance $d=66$, QAP obj=62]: the master played **cxd4**; under μ our *Pb2* plays the role of that game’s *Pc3*; cross-type roles: our *Pa3* as their *Rf1*, our *Be2* as their *Pe2*; transferred move **b2c3** (**endorses this move**). Synthesis: 8 retrieved analogs transferred to **bxc3** (summed QAP objective 407, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



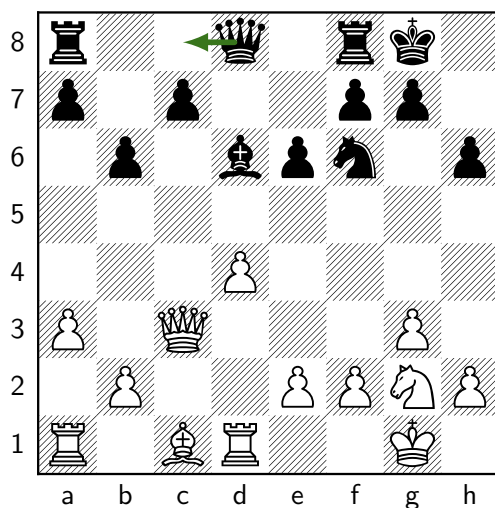
(a)



(b)

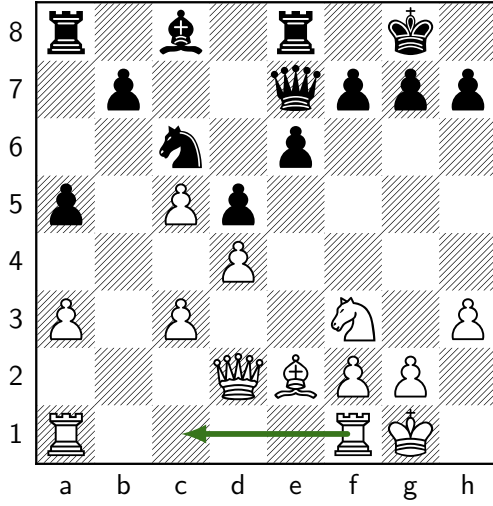


(c)

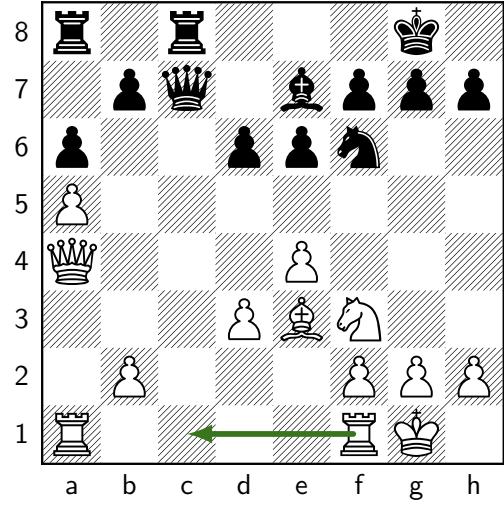


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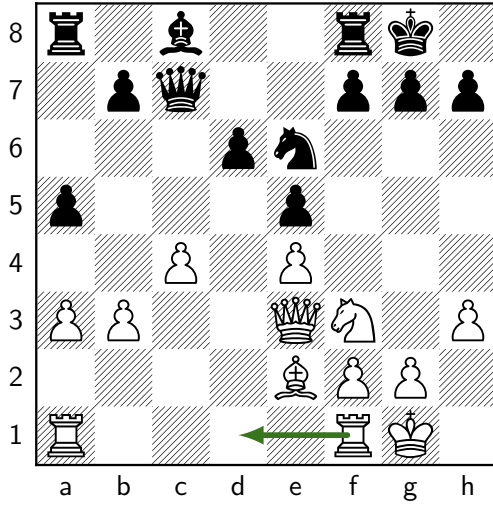
Figure 10: **Move 16, White: Qd2.** (a) Our position; Melanie plays Qd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#551` (ply 27) [structural distance $d=53$, QAP obj=47]: the master played Qe7; under μ our Qc1 plays the role of that game's Qd1; cross-type roles: our bc8 as their Pg7, our nc6 as their Bd6; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#552` (ply 27) [structural distance $d=53$, QAP obj=47]: the master played Qe7; under μ our Qc1 plays the role of that game's Qd1; cross-type roles: our bc8 as their Pg7, our nc6 as their Bd6; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#354` (ply 27) [structural distance $d=53$, QAP obj=47]: the master played Qc8; under μ our Qc1 plays the role of that game's Qd1; cross-type roles: our bc8 as their Pg7, our nc6 as their Bd6; transferred move c1d2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qd2 (summed QAP objective 262, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



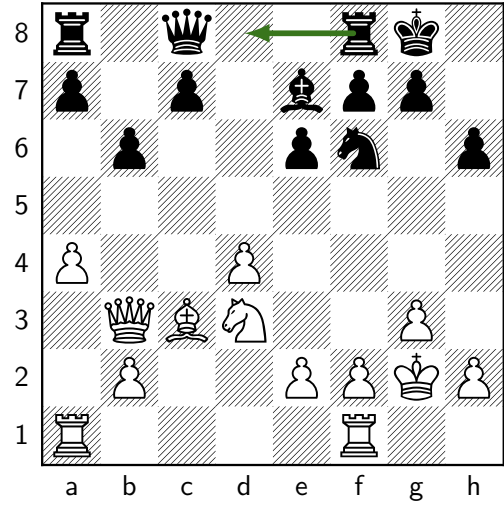
(a)



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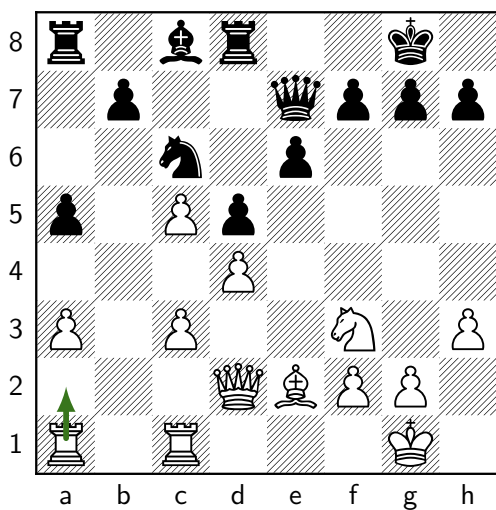


(c)

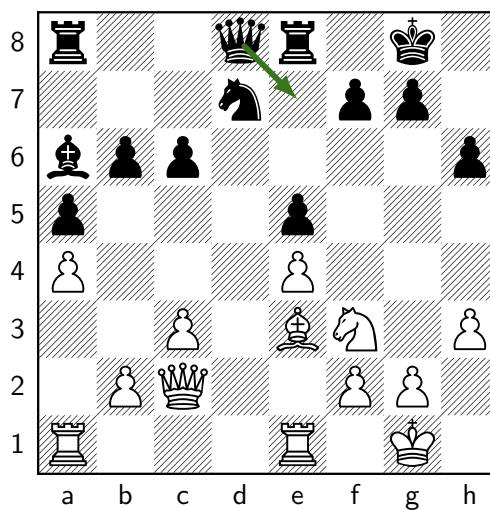


(d)

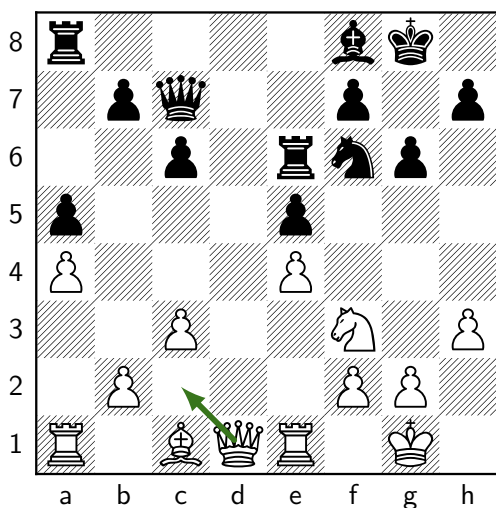
Figure 11: **Move 17, White: Rfc1.** (a) Our position; Melanie plays Rfc1 (f1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#744` (ply 30) [structural distance $d=50$, QAP obj=47]: the master played Rfc1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Be2* as their *Pg2*, our *Qd2* as their *Ph2*; transferred move f1c1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3430` (ply 32) [structural distance $d=57$, QAP obj=39]: the master played Rfd1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pe6* as their *Ra1*, our *Nf3* as their *Pa3*; transferred move f1c1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3257` (ply 31) [structural distance $d=58$, QAP obj=35]: the master played Rd8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Ph3* as their *Ra8*, our *Ra1* as their *Qc8*; transferred move f1c1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rfc1 (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



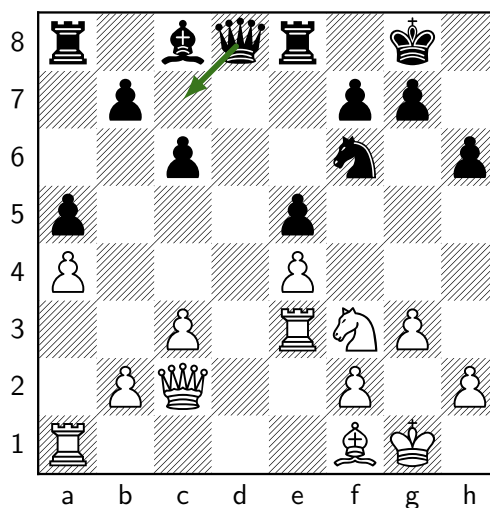
(a)



(b)

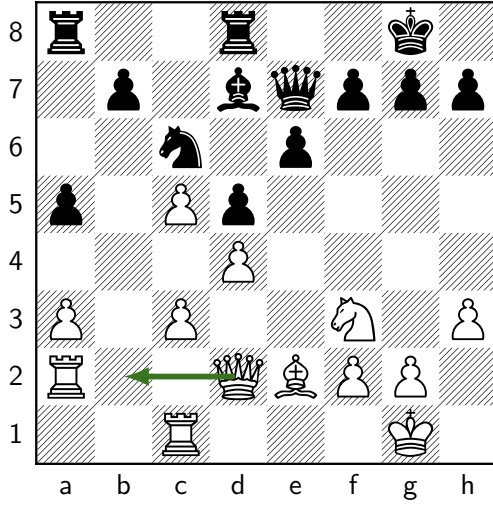


(c)

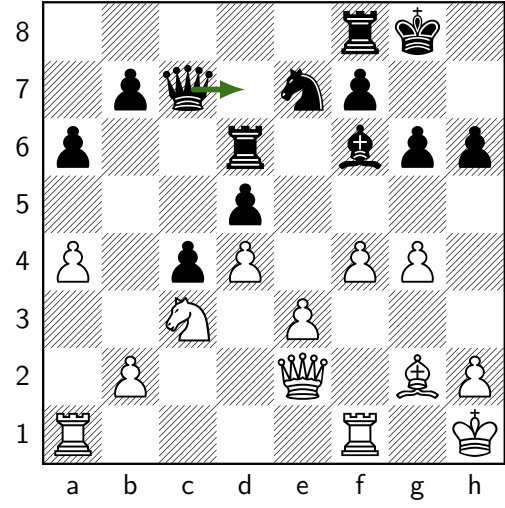


(d)

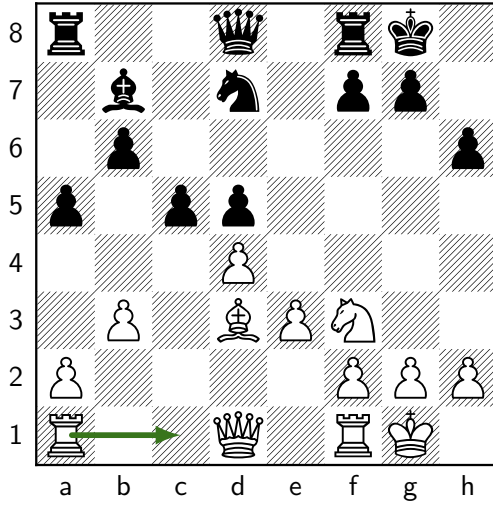
Figure 12: **Move 18, White: Ra2.** (a) Our position; Melanie plays Ra2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3942` (ply 33) [structural distance $d=68$, QAP obj=49]: the master played Qe7; under μ our Ra1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ra8, our Nf3 as their Re8; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1420` (ply 32) [structural distance $d=56$, QAP obj=45]: the master played Qc2; under μ our Ra1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pe6 as their Ra1, our Qd2 as their Bc1; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#909` (ply 31) [structural distance $d=56$, QAP obj=45]: the master played Qc7; under μ our Ra1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pe6 as their Ra8, our Qd2 as their Bc8; transferred move a1a2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ra2 (summed QAP objective 196, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



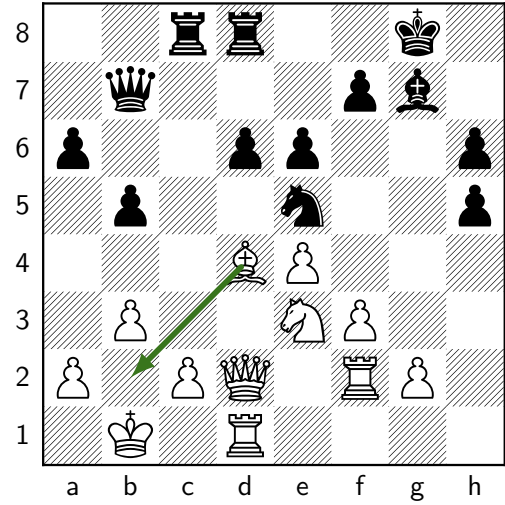
(a)



(b)

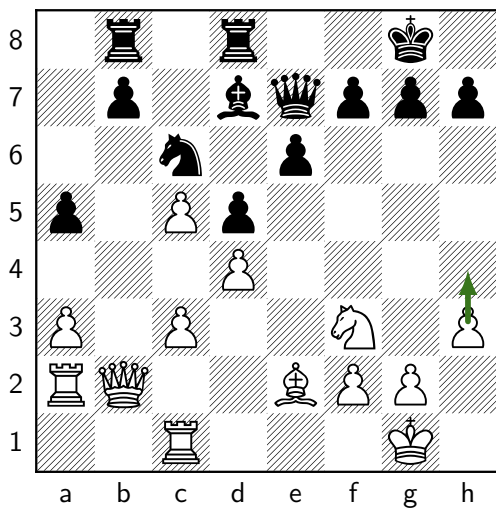


(c)

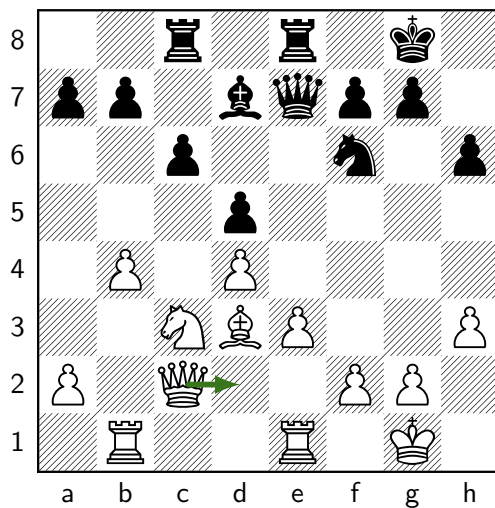


(d)

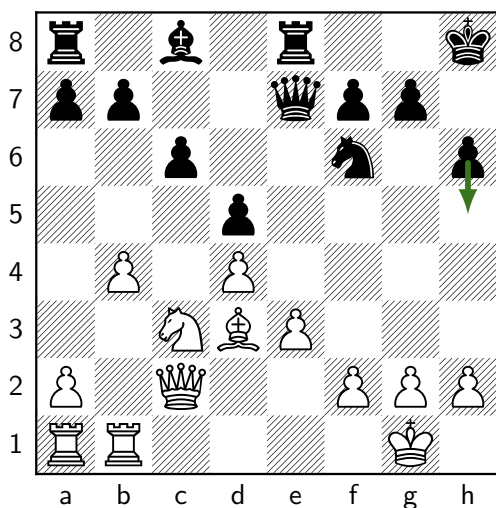
Figure 13: **Move 19, White: Qb2.** (a) Our position; Melanie plays Qb2 (d2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2663` (ply 39) [structural distance $d=52$, QAP obj=69]: the master played Qd7; under μ our Qd2 plays the role of that game’s Qc2; cross-type roles: our Nf3 as their Rf8, our pe6 as their Ne7; transferred move d2b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#860` (ply 28) [structural distance $d=56$, QAP obj=16]: the master played Rc1; under μ our Qd2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Be2 as their Qd1, our Pf2 as their Rf1; transferred move d2b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2042` (ply 44) [structural distance $d=53$, QAP obj=11]: the master played Bb2; under μ our Qd2 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Rd1, our Ra2 as their Pc2; transferred move d2b2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qb2 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



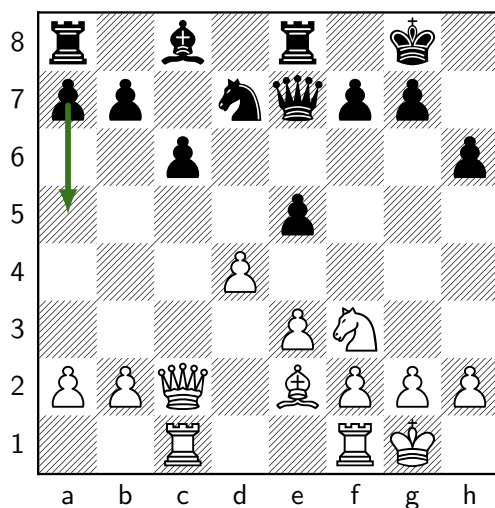
(a)



(b)

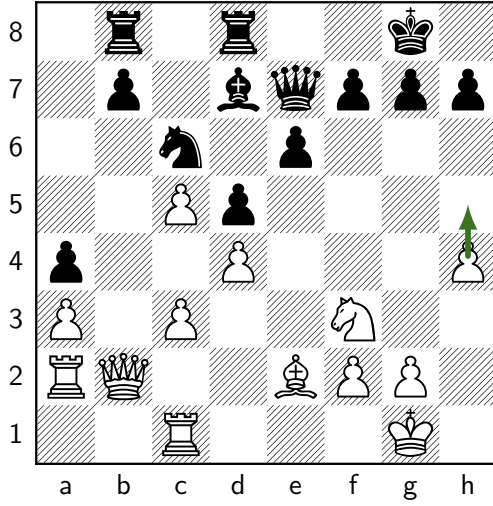


(c)

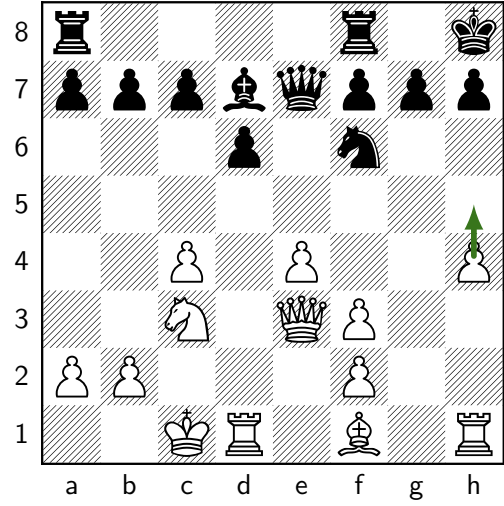


(d)

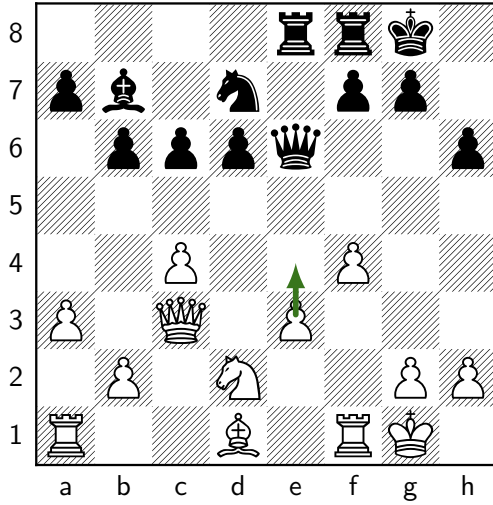
Figure 14: **Move 20, White: h4.** (a) Our position; Melanie plays h4 (h3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2942` (ply 38) [structural distance $d=60$, QAP obj=74]: the master played Qd2; under μ our *Ph3* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Rc1* as their *Pa2*, our *Nf3* as their *Pg2*; transferred move h3h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1199` (ply 33) [structural distance $d=54$, QAP obj=47]: the master played h5; under μ our *Ph3* plays the role of that game’s *Ph3*; cross-type roles: our *Qb2* as their *Ra8*, our *Ra2* as their *Bc8*; transferred move h3h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3308` (ply 29) [structural distance $d=55$, QAP obj=43]: the master played a5; under μ our *Ph3* plays the role of that game’s *Pa2*; cross-type roles: our *Pc5* as their *Ra8*, our *rb8* as their *Bc8*; transferred move h3h4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to h4 (summed QAP objective 262, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



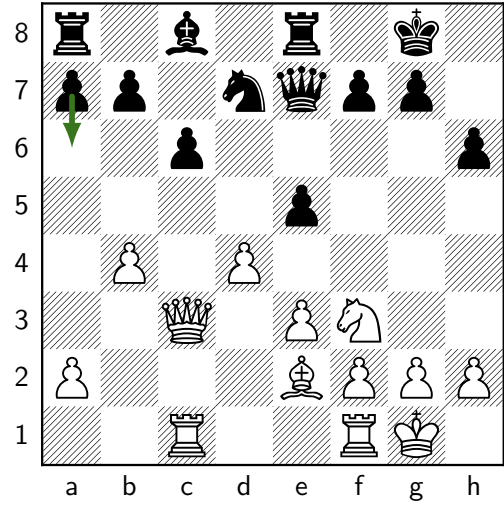
(a)



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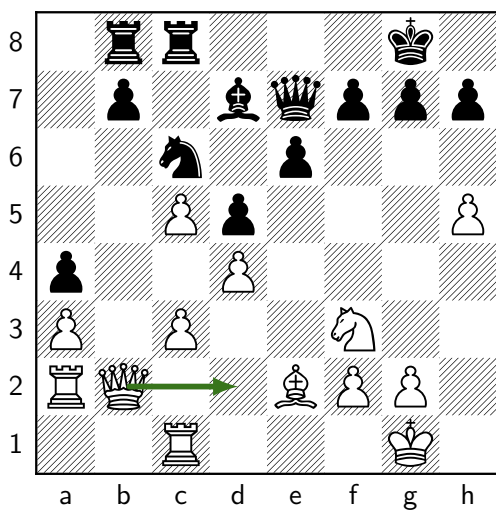


(c)

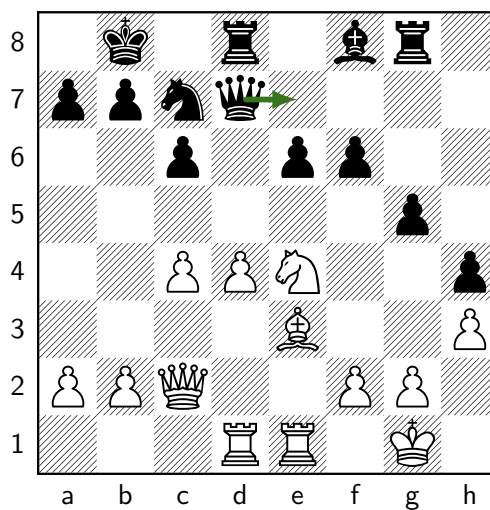


(d)

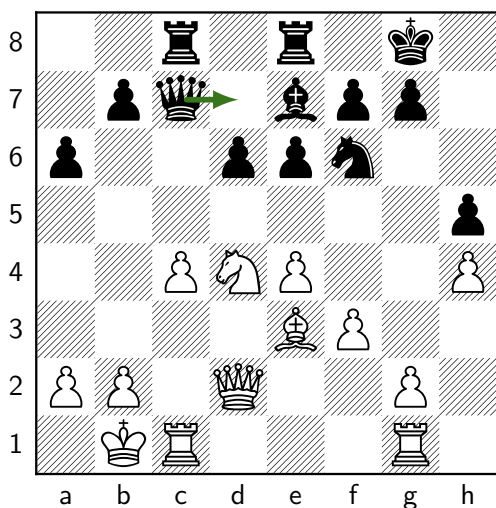
Figure 15: **Move 21, White: h5.** (a) Our position; Melanie plays h5 (h4h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3464` (ply 26) [structural distance $d=54$, QAP obj=60]: the master played h5; under μ our Ph_4 plays the role of that game’s Ph_4 ; cross-type roles: our $Qb2$ as their $Rd1$, our $Rc1$ as their $Bf1$; transferred move h4h5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1155` (ply 36) [structural distance $d=53$, QAP obj=46]: the master played e4; under μ our Ph_4 plays the role of that game’s $Pe3$; cross-type roles: our $Qb2$ as their $Ra1$, our $Nf3$ as their $Bd1$; transferred move h4h5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#70` (ply 29) [structural distance $d=61$, QAP obj=41]: the master played a6; under μ our Ph_4 plays the role of that game’s $Pa2$; cross-type roles: our $Pf2$ as their $Ra8$, our $Pc3$ as their $Bc8$; transferred move h4h5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to h5 (summed QAP objective 222, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



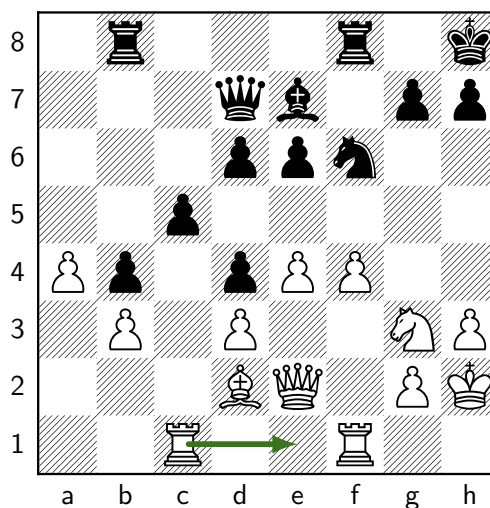
(a)



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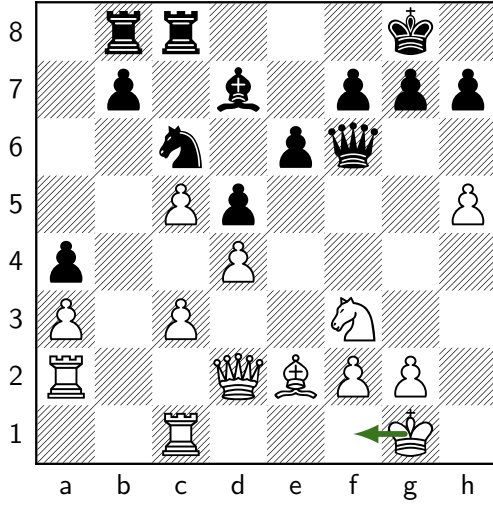


(c)

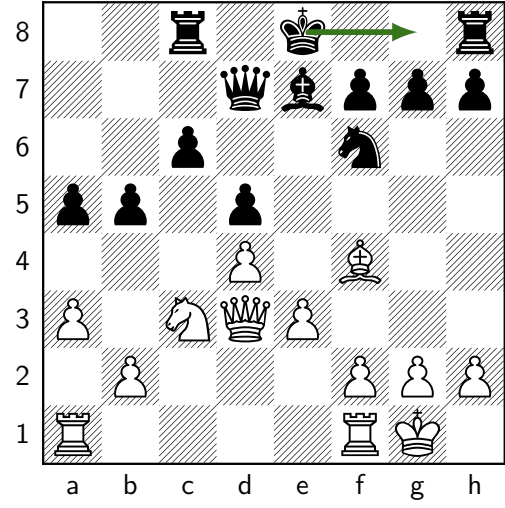


(d)

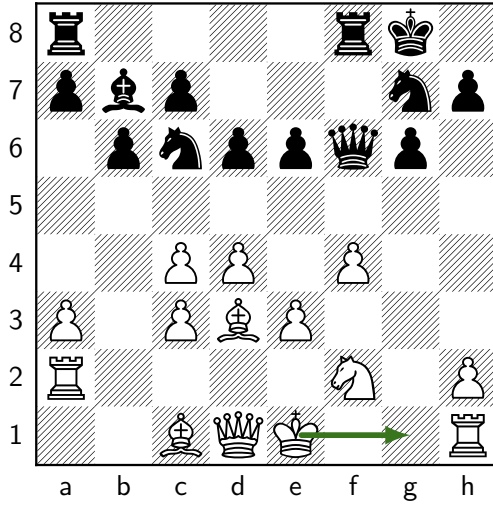
Figure 16: **Move 22, White: Qd2.** (a) Our position; Melanie plays Qd2 (b2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3988` (ply 35) [structural distance $d=61$, QAP obj=83]: the master played Qe7; under μ our Qb2 plays the role of that game’s Qd2; cross-type roles: our Nf3 as their Rd8, our Pc5 as their Bf8; transferred move b2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2670` (ply 37) [structural distance $d=47$, QAP obj=70]: the master played Qd7; under μ our Qb2 plays the role of that game’s Qc2; cross-type roles: our Nf3 as their Re8, our Rc1 as their Be7; transferred move b2d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#86` (ply 50) [structural distance $d=60$, QAP obj=47]: the master played Rce1; under μ our Qb2 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our pa4 as their Rf1, our nc6 as their Bd2; transferred move b2d2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qd2 (summed QAP objective 201, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



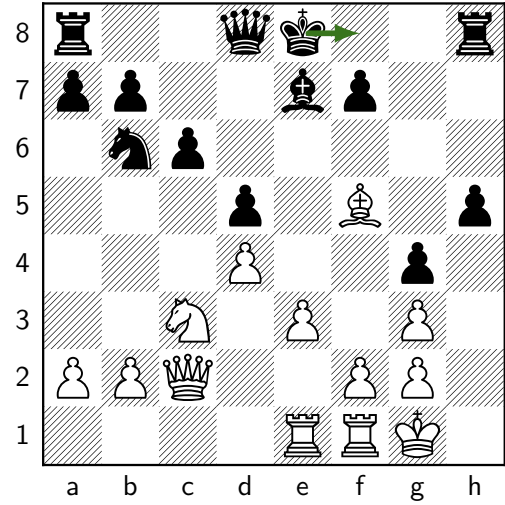
(a)



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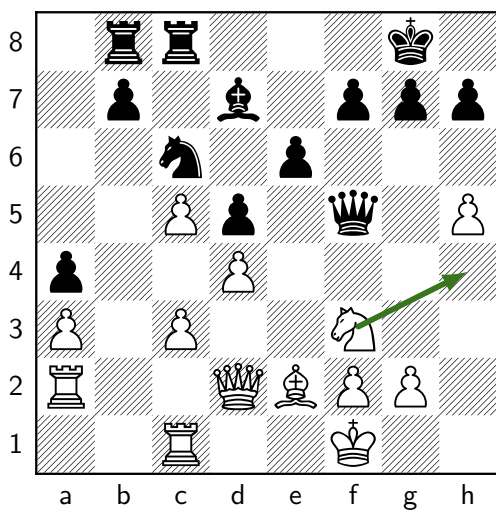


(c)

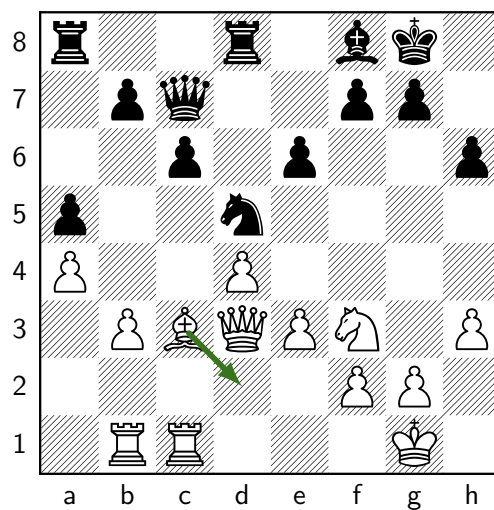


(d)

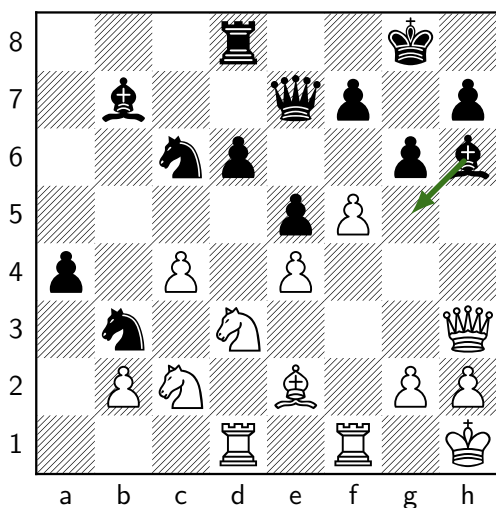
Figure 17: **Move 23, White: Kf1**. (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3333` (ply 29) [structural distance $d=56$, QAP obj=43]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pc5* as their *Rc8*, our *Pf2* as their *Rh8*; transferred move g1f1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3835` (ply 32) [structural distance $d=66$, QAP obj=36]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd2* as their *Bc1*, our *pg7* as their *Qd1*; transferred move g1f1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#269` (ply 33) [structural distance $d=59$, QAP obj=32]: the master played Kf8; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd2* as their *Ra8*, our *Rc1* as their *Qd8*; transferred move g1f1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf1 (summed QAP objective 141, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



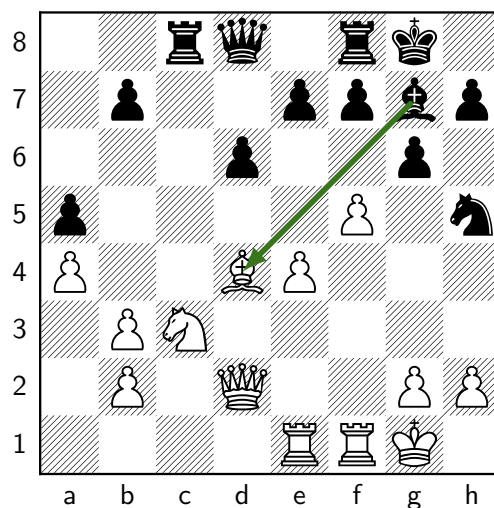
(a)



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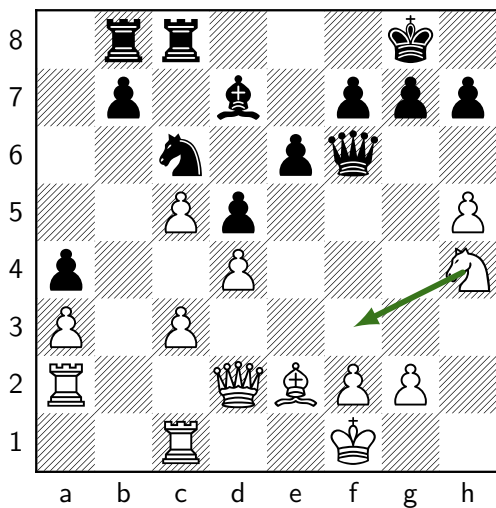


(c)

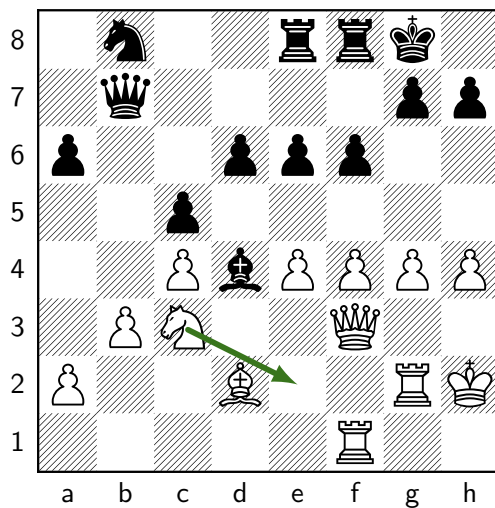


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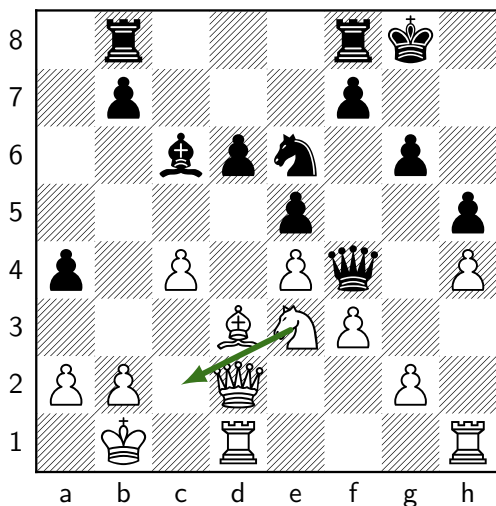
Figure 18: **Move 24, White: Nh4.** (a) Our position; Melanie plays Nh4 (f3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1098` (ply 56) [structural distance $d=69$, QAP obj=37]: the master played Bd2; under μ our Nf3 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Be2 as their Rc1, our bd7 as their Qd3; transferred move f3h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#861` (ply 57) [structural distance $d=72$, QAP obj=36]: the master played Bg5; under μ our Nf3 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Bb7, our Pc5 as their Nc6; transferred move f3h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2097` (ply 31) [structural distance $d=71$, QAP obj=32]: the master played Bxd4+; under μ our Nf3 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our nc6 as their Rc8, our bd7 as their Qd8; transferred move f3h4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Nh4 (summed QAP objective 184, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



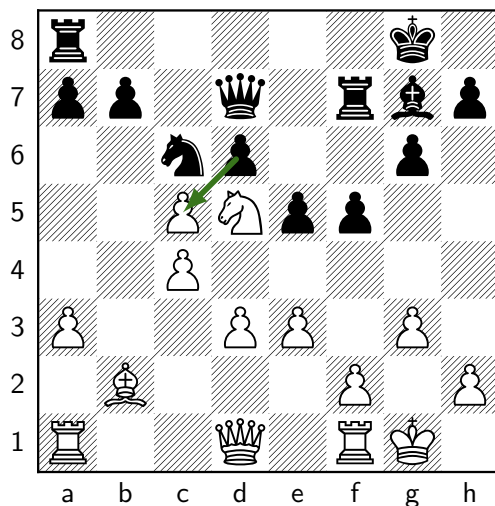
(a)



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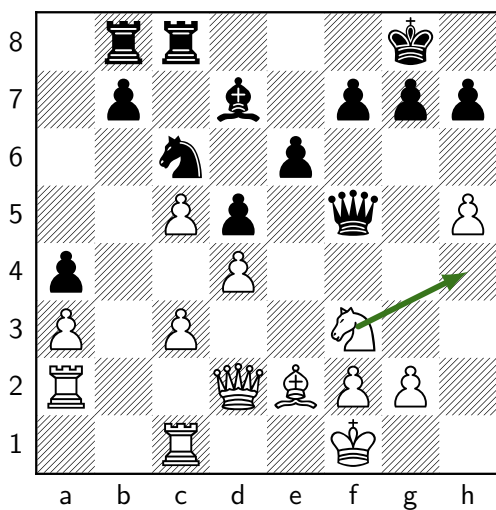


(c)

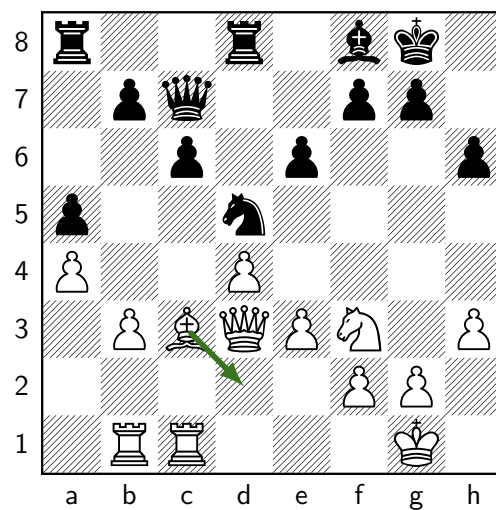


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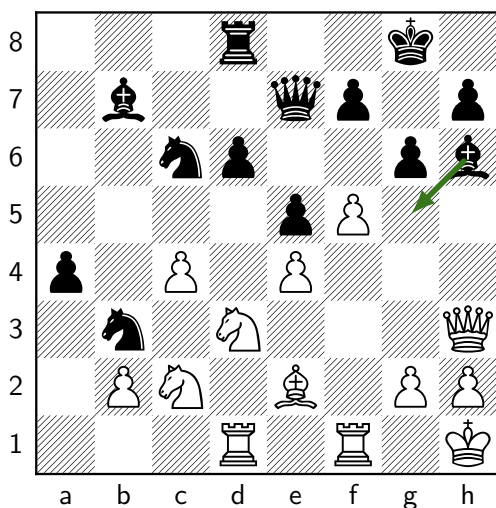
Figure 19: **Move 25, White: Nf3.** (a) Our position; Melanie plays Nf3 (h4f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3997` (ply 52) [structural distance $d=67$, QAP obj=37]: the master played Ne2; under μ our Nh4 plays the role of that game’s Nc3; cross-type roles: our Pg2 as their Rf1, our Qd2 as their Bd2; transferred move h4f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1919` (ply 40) [structural distance $d=71$, QAP obj=34]: the master played Nc2; under μ our Nh4 plays the role of that game’s Ne3; cross-type roles: our Be2 as their Rd1, our Qd2 as their Rh1; transferred move h4f3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#21` (ply 33) [structural distance $d=77$, QAP obj=30]: the master played dxc5; under μ our Nh4 plays the role of that game’s Pd3 (a **cross-type** correspondence); cross-type roles: our Qd2 as their Ra8, our Be2 as their Qd7; transferred move h4f3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nf3 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



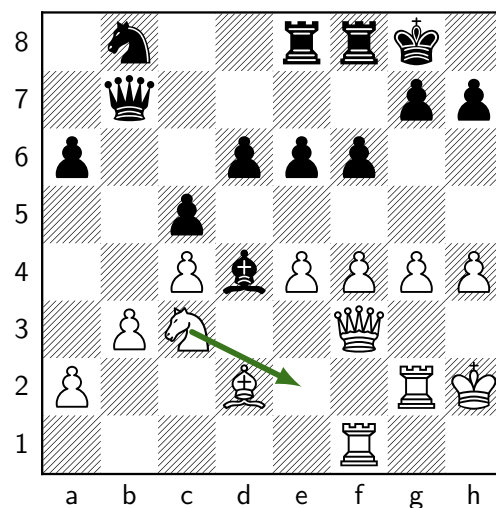
(a)



(b)



(c)



(d)

Figure 20: **Move 26, White: Nh4.** (a) Our position; Melanie plays Nh4 (f3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1098` (ply 56) [structural distance $d=69$, QAP obj=37]: the master played Bd2; under μ our Nf3 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Be2 as their Rc1, our bd7 as their Qd3; transferred move f3h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#861` (ply 57) [structural distance $d=72$, QAP obj=36]: the master played Bg5; under μ our Nf3 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Bb7, our Pc5 as their Nc6; transferred move f3h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3997` (ply 52) [structural distance $d=64$, QAP obj=17]: the master played Ne2; under μ our Nf3 plays the role of that game’s Nc3; cross-type roles: our Pc5 as their Rf1, our Be2 as their Rg2; transferred move f3h4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nh4 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.